

MATH 263 Notes

Chapter 3 - The Logic of Quantified Statements

Natalie Piltoyan

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Introduction

Before we start this chapter, here is some relevant notation:

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ - the set of natural numbers
- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ - the set of integers
- $\mathbb{Q} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$ - the set of rational numbers
- $\mathbb{R}$ - the set of real numbers

3.1 - Predicates and Quantified Statements I

Vaguely speaking, a *predicate* is a sentence that contains a finite number of variable(s) and may be true or false depending on the values of the variable(s).

Consider the sentence “Sulaimon is a student at Towson University.” If we let P represent “is a student at Towson University” and Q represent “Sulaimon”. Then P and Q are called *predicate symbols*. If, however, we replace “Sulaimon” with a variable x that takes in values, then the sentence becomes “ x is a student at Towson University” and can be symbolized as $P(x)$.

In that example, x is called a *predicate symbol* and “ x is a student at Towson University” is a predicate.

The *domain* of a predicate variable is the set of all values that may be substituted in place of the variable.

The *truth set* of a predicate is the set of all elements in the domain that make the predicate true.

Up until now, we have limited our types of statements to conditional: if p , then q . For words like “some” and “all”, we use *quantifiers*.

The *universal quantifier* is denoted with $\forall$. It is often read as “for all” or “for each.” For example, the statement

$$\forall x \in \mathbb{N}, x \geq 0$$

is read as “for all natural numbers x , x is greater than or equal to 0” (which is true!).

Let $P(x)$ be a predicate and D be the domain of x . A *universal statement* is a statement of the form $\forall x \in D, P(x)$. To prove statements like these, you have to show that for **every single** x in the domain D , $P(x)$ is true. If there is a value for x for which $P(x)$ is false, then the statement is false, and x is a *counterexample*. For smaller domains, you can show that a predicate is true for every single value (this is called the *method of exhaustion*. For larger domains, other techniques are needed.

The *existential quantifier* is denoted with $\exists$. It is often read as “there exists” or “there is at least one.” The statement

$$\exists x \in \mathbb{N} \text{ s.t. } x > 5$$

is read as “there exists a natural number x such that x is greater than 5.”

Similarly to universal statements, an *existential statement* is a statement of the form $\exists x \in D$ such that $P(x)$. For these, the statement is true if and only if $P(x)$ is true for at least one value of $x \in D$. If it is false for every single value of $x \in D$, then $P(x)$ is false.

Implications using Predicates

Let $P(x)$ and $Q(x)$ be predicates and let D be the overlapping domain.

- The statement $P(x) \implies Q(x)$ means that for every element $x \in D$, every truth value in the truth set of $P(x)$ is also in the truth set of $Q(x)$. In notation: $\forall x \in D, P(x) \implies Q(x)$.
- The statement $P(x) \iff Q(x)$ means that for every element $x \in D$, the truth values in the truth set of $P(x)$ and $Q(x)$ are identical. In notation: $\forall x \in D, P(x) \iff Q(x)$.

3.2 - Predicates and Quantified Symbols II

The negation of a universal statement $\forall x \in D, P(x)$ is $\exists \in D$ s.t. $\neg P(x)$. So,

$$\neg(\forall x \in D, P(x)) \equiv \exists \in D \text{ s.t. } \neg P(x)$$

The negation of an existential statement $\exists x \in D$ such that $P(x)$ is $\forall \in D, \neg P(x)$. So,

$$\neg(\exists \in D \text{ s.t. } P(x)) \equiv \forall x \in D, \neg P(x).$$

The negation of a universal conditional statement $\forall x, P(x) \implies Q(x)$ is $\exists x$ such that $\neg(P(x) \implies Q(x))$. Since the negation of $P(x) \implies Q(x)$ is $P(x) \wedge \neg Q(x)$, we have

$$\neg(\forall x, P(x) \implies Q(x)) \equiv \exists x \text{ s.t. } P(x) \wedge \neg Q(x).$$

Our definitions of contrapositive, converse, and inverse do not change much. Given a universal conditional statement $\forall x \in D, P(x) \implies Q(x)$,

- Contrapositive: $\forall x \in D, \neg Q(x) \implies \neg P(x)$.
- Converse: $\forall x \in D, Q(x) \implies P(x)$.
- Inverse: $\forall x \in D, \neg P(x) \implies \neg Q(x)$.

3.3 - Statements with Multiple Quantifiers

So far, we have only seen statements with one quantifier. In this section, we will see sentences with multiple quantifiers.

Important Note - The order in which the quantifiers are read matters!

Let D and E be domains, $x \in D$, $y \in E$, and $P(x, y)$ be a predicate.

If you have a statement such as

$$\forall x \in D, \exists y \in E \text{ s.t. } P(x, y),$$

then your goal is to show that, for every single x , there is a y that satisfies $P(x, y)$.

If you have a statement such as

$$\exists x \in D \text{ s.t. } \forall y \in E, P(x, y),$$

then your goal is to show that there is at least one x that works with every single y .

To take the negation of a statement with multiple quantifiers, switch the first quantifier and then take the negation of the rest of the statement.

$$\begin{aligned} \neg(\forall x \in D, \exists y \in E \text{ s.t. } P(x, y)) &\equiv \exists x \in D \text{ s.t. } \neg(\exists y \in E \text{ s.t. } P(x, y)) \\ &\equiv \exists x \in D \text{ s.t. } \forall y \in E, \neg P(x, y). \end{aligned}$$

$$\begin{aligned} \neg(\exists x \in D \text{ s.t. } \forall y \in E, P(x, y)) &\equiv \forall x \in D, \neg(\forall y \in E, P(x, y)) \\ &\equiv \forall x \in D, \exists y \in E \text{ s.t. } \neg P(x, y). \end{aligned}$$